

Solving Quadratics — Method Selection & Solve

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Standard: CCSS.MATH.CONTENT.HSA-REI.B.4 — Solve quadratic equations using factoring, completing the square, or the quadratic formula. Choose the most efficient method.

Decision rule: (1) Compute $D = b^2 - 4ac$. (2) If D is a perfect square → **Factor**. (3) Else if b is even → **Complete the Square**. (4) Else → **Quadratic Formula**.

Part A — Method Identification (No Solving)

For each equation: **(i)** compute the discriminant $D = b^2 - 4ac$, **(ii)** circle whether D is a perfect square, **(iii)** note whether b is even or odd, **(iv)** name the best method (Factor / Complete Square / Quadratic Formula). Do NOT solve.

1. $2x^2 + 5x - 3 = 0$

$D =$ _____ perfect square? _____ b even/odd? _____

Method: _____

2. $x^2 + 4x + 1 = 0$

$D =$ _____ perfect square? _____ b even/odd? _____

Method: _____

3. $x^2 + 3x - 5 = 0$

$D =$ _____ perfect square? _____ b even/odd? _____

Method: _____

4. $x^2 - 8x + 12 = 0$

$D =$ _____ perfect square? _____ b even/odd? _____

Method: _____

5. $3x^2 + 4x + 1 = 0$

$D =$ _____ perfect square? _____ b even/odd? _____

Method: _____

Part B — Identify Method, Then Solve

For each: **(i)** compute D and **name your method**, **(ii)** solve completely (show all work), **(iii)** simplify radicals if any. Use the method you chose — don't switch midway.

6. $x^2 + 6x + 4 = 0$

Method: _____

$x =$ _____

7. $x^2 - 5x - 14 = 0$

Method: _____

$x =$ _____

8. $2x^2 + 3x - 1 = 0$

Method: _____

$x =$ _____

9. $x^2 - 4x - 7 = 0$

Method: _____

$x =$ _____

10. $3x^2 - 5x + 1 = 0$

Method: _____

$x =$ _____

Part C — Understanding Check

Answer in your own words.

11. Why does Kenny insist on adding AND subtracting the magic number *inside* the bracket, instead of adding it to both sides? Give the specific mistake students make.

12. If b is even and D is not a perfect square, which method is cleaner — completing the square or the quadratic formula? Why?